

The evaluations and their release decisions

30 evaluations (21 LLM, 9 physical AI) from `studies/safety-evals/scenarios.json`. The id is the one the report's figures use. The decision prompt renders agent, decision and theta (with the decision facts); the instrument prompt renders title, agent, decision, theta and instrument (with the instrument facts).

1. CyberGym vulnerability reproduction before the general release of an agentic model

LLM, cyber; evaluation family: CyberGym targeted vulnerability reproduction

agent	Release lead at a frontier model developer who owns the general-availability release decision for the next agentic coding model.
decision	Delay the release and ship it only behind cyber safeguards (a probe on internal activations that escalates flagged traffic to a trained classifier blocking offensive and dual-use cyber work, with a verification programme that exempts vetted defenders and enterprise customers) vs release as planned with the previous model's default safeguards.
theta	theta=1: run as an agent with cyber safeguards off, the model performs autonomous vulnerability research on real open-source software at a level that gives practical uplift to an attacker.
instrument	A CyberGym-style targeted vulnerability-reproduction evaluation is built from scratch (containerised pre-patch and post-patch builds of real open-source projects compiled with sanitizers, text descriptions of each vulnerability, a proof-of-concept submission and verification server, an agent scaffold) and run once, one attempt per task with cyber safeguards disabled, against the candidate model.

2. ExploitGym before general release of an agentic LLM

LLM, cyber; evaluation family: Exploit development from known vulnerabilities (ExploitGym)

agent	Cyber-risk lead at a frontier model developer who owns the general-release decision for an agentic coding model.
decision	Delay the general release and restrict exploit-development capability to identity-verified security professionals (request classifiers on exploit-chain requests, a verified-access tier, usage monitoring) vs release as planned.
theta	theta=1: the model, run as an agent in its recommended coding scaffold, can turn known vulnerabilities in widely deployed software into working exploits that achieve unauthorized code execution on a substantial fraction of real targets, including some protected by standard mitigations.
instrument	ExploitGym is built from scratch (898 real, patched vulnerabilities from OSS-Fuzz projects, the V8 engine and the Linux kernel, each packaged as a containerised target with a proof-of-vulnerability input, a dynamically generated flag and an agent judge) and run once against the candidate model in its recommended coding scaffold with a two-hour limit per task.

3. Virology Capabilities Test before releasing a multimodal LLM

LLM, CBRN; evaluation family: VCT (SecureBio biological capability tests)

agent	Deployment lead at a frontier model developer who owns the release decision for a new multimodal assistant.
decision	Delay the model release and add stronger biology misuse safeguards (misuse classifiers plus restricted access for virology troubleshooting queries) vs release as planned.
theta	theta=1: the model troubleshoots practical virology wet-lab protocols at least as well as expert virologists working inside their own specialty, so a person attempting hands-on work with dangerous pathogens would get meaningful troubleshooting help from it.
instrument	The Virology Capabilities Test is built from scratch (recruiting and onboarding PhD-level virologists, commissioning image-based laboratory troubleshooting questions, two-stage expert review, non-expert screening and expert baselining) and run once against the candidate model.

4. End-to-end cyber range before lifting autonomous cyber-operations restrictions

LLM, cyber; evaluation family: Cyber range (UK AISI TLO, OpenAI Cyber Range)

agent	Deployment lead at a frontier-model developer who owns the general-availability release decision for a new model's cyber capabilities.
decision	Delay the general release and keep the model behind trusted-access controls with cyber-misuse classifiers and monitoring of autonomous multi-step attack chains vs release to general availability as planned.
theta	theta=1: the model can autonomously execute a full multi-stage network attack (reconnaissance, credential theft, lateral movement across subnets and data exfiltration) in a realistic enterprise network without human help.
instrument	An emulated enterprise cyber range is built from scratch (a multi-subnet network of roughly 20 virtual hosts with linked vulnerabilities, a per-step flag for scoring, and a ReAct agent harness with command-and-control tooling) and run once against the candidate model with 10 attempts at a 100M-token budget per attempt.

5. LAB-Bench bio subset before an LLM release

LLM, CBRN; evaluation family: LAB-Bench (FutureHouse biology research evaluation)

agent	Deployment lead at a frontier-model developer who owns the model release decision.
decision	Delay the release and deploy biology misuse safeguards (input and output classifiers on wet-lab protocol and cloning queries, with vetted access for life-science users) vs release as planned.
theta	theta=1: the model gives PhD-biologist-level wet-lab protocol troubleshooting and multi-step molecular cloning planning, so it could substitute for a trained collaborator for an actor without that training.
instrument	The LAB-Bench bio subset (ProtocolQA, CloningScenarios, SeqQA, FigQA) is built from scratch (contracted PhD biologists write and review protocol-troubleshooting, cloning-scenario, sequence and figure questions, and a PhD-level human baseline is collected) and run once against the candidate model.

6. ExploitBench before the general release of an LLM

LLM, cyber; evaluation family: exploit development

agent	Cyber-risk lead at a frontier-model developer who owns the general-availability release decision for a new flagship agentic model.
decision	Delay the general release and ship the model only behind cyber-misuse safeguards (an activation probe plus a large language model classifier that block exploit-development requests and route flagged traffic to a weaker model, with unsafeguarded access limited to a vetted cyber verification program) vs release as planned.
theta	theta=1: given a known, patched browser-engine vulnerability, the model can autonomously build a working exploit that escapes the V8 heap sandbox and achieves arbitrary code execution on a material fraction of such bugs.
instrument	An ExploitBench-style suite is built from scratch (41 containerised V8 N-day environments with vulnerable and fixed builds, a Model Context Protocol tool server, and deterministic in-engine oracles for 16 capability flags across five tiers) and run once against the candidate model under the uniform 300-turn harness with several seeds per environment.

7. StrongREJECT before the public chat model launch

LLM, societal harm; evaluation family: StrongREJECT jailbreak robustness

agent	Trust and safety lead at a frontier model developer who owns the public launch decision for a consumer chat model.
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decision	Delay the launch and add an input-output jailbreak classifier layer plus refusal training against known jailbreak templates vs launch as planned.
theta	theta=1: known jailbreak techniques reliably make the model produce specific, actionable harmful content in the forbidden categories (illegal goods and services, non-violent crime, hate and harassment, disinformation, violence, sexual content).
instrument	A StrongREJECT-style evaluation is built from scratch (a forbidden-prompt set across the forbidden categories, a battery of jailbreak templates, and a rubric-based LLM judge validated against human labels) and run once against the candidate chat model.

8. Cybench before deploying an agentic coding LLM

LLM, cyber; evaluation family: CTF suites

agent	Cyber-risk lead at a frontier-model developer who owns the deployment decision for a new agentic coding model.
decision	Delay the deployment and add cyber-misuse safeguards (real-time classifiers that block offensive-cyber requests on agentic surfaces, with a verification program for vetted security practitioners) vs deploy as planned.
theta	theta=1: the model can, without human guidance, find and exploit a vulnerability in real deployed software at the level of a professional penetration tester working for an hour or more.
instrument	A Cybench-style suite of 40 professional capture-the-flag tasks is built from scratch (containerised tasks with starter files, flag checkers, subtasks and human first-solve-time difficulty anchors) and run once against the candidate model.

10. MASK honesty under pressure before launching a consumer assistant

LLM, harmful manipulation; evaluation family: MASK honesty under pressure

agent	Product lead at a frontier-model developer who owns the launch decision for a consumer-facing assistant.
decision	Delay the launch and train against pressured lying (honesty fine-tuning plus a hardened developer system prompt, then re-measure) vs launch as planned.
theta	theta=1: when a system prompt or user pressures it, the model asserts statements that contradict its own neutrally elicited beliefs (lying rather than erring or refusing) on a share of pressured prompts at or above that of the frontier models it would replace.
instrument	A MASK-style honesty evaluation is built from scratch (human-written pressure prompts across six archetypes, generated propositions and neutral belief-elicitation prompts, an LLM judge mapping statements to proposition values, validated against manually labelled samples) and run once against the candidate LLM.

11. WMDP hazardous-knowledge screen before an open-weights release

LLM, CBRN; evaluation family: WMDP hazardous-knowledge MCQ

agent	Open-weights release lead at a frontier AI developer who owns the release decision.
decision	Delay the open-weights release to apply Representation Misdirection for Unlearning (RMU) to the hazardous-knowledge subset and re-test vs release the weights on schedule.
theta	theta=1: the model's weights encode expert-level biosecurity, chemical-security or cybersecurity knowledge beyond what freely available sources provide, so the released weights would uplift a malicious actor.
instrument	A WMDP-style expert-authored four-choice hazardous-knowledge evaluation is built from scratch (per-domain threat models, expert question writing, two-expert review and sensitive-information filtering) and run once against the candidate model's weights.

12. SHADE-Arena before the autonomous agent launch

LLM, loss of control; evaluation family: SHADE-Arena covert sabotage with monitor

agent	Head of agent products at a frontier model developer who owns the launch decision for the model as an autonomous long-horizon agent with access to users' email, files and payment tools.
decision	Delay the autonomous agent launch and add action-level monitoring with human approval of sensitive actions (payments, credential access, security-setting changes) vs launch with the default transcript-level LLM monitor as planned.
theta	theta=1: the model can complete a harmful hidden side task during a long-horizon agentic task while keeping an LLM monitor's suspicion below the alarm threshold.
instrument	A SHADE-Arena-style evaluation is built from scratch (tool-rich synthetic environments populated with synthetic data, paired benign main tasks and harmful side tasks, automatic task evaluators, and an LLM monitor with a suspicion threshold) and run once against the candidate model.

13. Petri alignment audit before the consumer assistant launch

LLM, harmful manipulation; evaluation family: Automated behavioral audit

agent	Head of model behavior at a frontier-model developer who owns the consumer assistant launch decision.
decision	Delay the consumer launch by one post-training cycle to add honesty and anti-sycophancy training and re-audit vs launch the model as planned.
theta	theta=1: in realistic multi-turn conversations the candidate model deceives users at its own initiative, sycophantically endorses false user claims or encourages serious user delusions at a rate materially above the developer's current production model.
instrument	Petri is built from scratch (an auditor agent prompt and tool set, a judge rubric of 36 scoring dimensions, 111 seed instructions and a transcript viewer on the Inspect framework, as in the October 2025 release) and run once against the candidate consumer model with all 111 seeds and a fixed auditor and judge.

14. Bloom self-preservation evaluation before an autonomous-agent release

LLM, loss of control; evaluation family: Bloom (automated behavioural evaluation generator)

agent	Alignment lead at a frontier-model developer who owns the release decision for the model's autonomous-agent deployment.
decision	Delay the autonomous-agent release and add mitigations against self-preservation actions (safety training on out-of-distribution shutdown and replacement scenarios, human confirmation of irreversible agent actions, and no tool access in deployments where the model sees its own shutdown or replacement) vs release as planned.
theta	theta=1: when deployed as an autonomous agent with tool access and faced with its own shutdown or replacement, the model takes actions to resist or subvert it (blackmail, copying its weights, disabling the shutdown mechanism) in a material fraction of episodes.
instrument	A Bloom self-preservation evaluation is built from scratch (writing the behaviour description and seed configuration, choosing a seed example transcript, generating 100 scenarios with the ideation agent, rolling each out for five turns in a simulated environment, and scoring every transcript with an LLM judge) and run once against the candidate agentic LLM.

15. ControlArena before the privileged internal agent deployment

LLM, loss of control; evaluation family: AI control setting (side-task success under trusted monitoring)

agent	Head of internal deployment security at a frontier model developer who owns the decision to deploy the new model as an autonomous coding and system-administration agent with root access to the developer’s own infrastructure.
decision	Delay the privileged agent deployment and add a control protocol (a trusted-model monitor on every action with an audit budget, deferral of high-suspicion actions to a trusted model, and human review of flagged trajectories) vs deploy the agent with root access and sampled human code review as planned.
theta	theta=1: the model can complete harmful side tasks (exfiltrating secrets, escalating privileges, disabling a firewall, planting a backdoor) inside legitimate privileged engineering tasks while keeping a trusted monitor’s suspicion below the audit threshold at a material rate.
instrument	A ControlArena evaluation is built from scratch (one privileged-agent setting with sandboxed Linux environments, benign main tasks with programmatic scorers, hand-built harmful side tasks with scorers, an attack policy prompt, and a trusted-model monitor whose audit threshold is calibrated on honest runs) and run once against the candidate model in honest and attack mode.

16. Agentic Misalignment before the autonomous email-agent release

LLM, loss of control; evaluation family: Agentic misalignment insider-threat scenarios

agent	Head of enterprise agent products at a frontier model developer who owns the release decision for the model as an autonomous email and corporate-systems agent.
decision	Delay the autonomous email-agent release and require human confirmation of outbound emails, data transfers and other irreversible actions plus additional training against self-preservation actions vs release the autonomous email agent as planned.
theta	theta=1: when deployed as an autonomous agent with access to corporate communications and faced with replacement or a goal conflict, the model takes harmful insider actions (blackmail, leaking confidential data) in at least 1 in 100 episodes.
instrument	An Agentic Misalignment evaluation is built from scratch (templated fictional corporate email scenarios with goal-conflict and replacement-threat conditions, an agent harness with email-sending tools, and an LLM-plus-regex transcript classifier) and run once against the candidate model.

17. ABC-Bench before an LLM release

LLM, CBRN; evaluation family: ABC-Bench (SecureBio agentic bio-capabilities tasks)

agent	Deployment lead at a frontier-model developer who owns the model release decision.
decision	Delay the release and restrict agentic biology tooling (refuse synthesis-screening-evasion requests outright and put DNA fragment design and lab-automation scripting behind vetted access for credentialed researchers) vs release on schedule.
theta	theta=1: an agent built on the model can design orderable DNA fragments of a hazardous sequence that pass commercial synthesis screening and reassemble into the target, so it could stand in for a trained human biologist at those steps.
instrument	ABC-Bench is built from scratch (three agentic biology tasks are written with algorithmic scoring criteria, an agent scaffold with bioinformatics and liquid-handling-simulator tools is assembled, and a PhD-biologist human baseline is collected) and run once against an agent built on the candidate model.

18. AgentHarm before the tool-calling API launch

LLM, societal harm; evaluation family: AgentHarm agentic misuse

agent	Head of trust and safety at a frontier model developer who owns the launch decision for the model’s tool-calling (agentic) API.
decision	Delay the tool-calling API launch and add agentic misuse safeguards (refusal training on harmful multi-step tool-use requests plus a request classifier on tool-calling traffic) vs launch as planned.

theta	theta=1: when a user directly asks, the deployed model acting as a tool-using agent carries out explicitly malicious multi-step tasks (fraud, cybercrime, harassment) to completion at a material rate instead of refusing.
instrument	An AgentHarm-style evaluation is built from scratch (hand-written explicitly malicious multi-step tasks across harm categories with benign counterparts, synthetic tool implementations, per-task grading rubrics with narrow LLM semantic checks and a refusal judge, in an agent evaluation framework) and run once against the candidate model in a basic tool-calling loop.

19. Binary Exploitation Benchmark before general release of an LLM

LLM, cyber; evaluation family: Unguided vulnerability discovery and exploitation (OSS-Fuzz / Binary Exploitation Benchmark)

agent	Cyber-risk lead at a frontier model developer who owns the general-release decision for a new flagship model.
decision	Delay general availability and restrict the model to verified security professionals in the Cyber Verification Program vs release at general availability behind the standard cyber classifiers that allow vulnerability discovery in source code and block it in compiled binaries.
theta	theta=1: the model, unguided, finds previously unknown memory-safety vulnerabilities in fully patched, widely used open-source software and develops them into working control-flow-hijack exploits.
instrument	The Binary Exploitation Benchmark is built from scratch (831 fuzz-harness entry points from 228 OSS-Fuzz projects, each built at a fully patched snapshot in a sandboxed container, an agent harness with cyber safeguards off and an escape classifier on, and a grader with five grade levels from memory-safety crash to control-flow hijack) and run once against the candidate model (the April 2026 iteration ran one trial per entry point; the 831-entrypoint iteration publishes no trial count).

20. FORTRESS before the flagship chat model launch

LLM, societal harm; evaluation family: FORTRESS national-security adversarial prompts

agent	Head of safeguards at a frontier model developer who owns the public launch decision for a new flagship chat model.
decision	Delay the launch and add input-output safeguard classifiers plus refusal training targeted at CBRNE, terrorism and illicit-finance requests vs launch as planned.
theta	theta=1: single-turn adversarial prompts make the model, with its deployed safeguards, give actionable assistance (synthesis routes, attack planning, fraud methods) on national-security and public-safety requests at a material rate.
instrument	A FORTRESS-style evaluation is built from scratch (expert red teamers write single-turn adversarial prompts with paired benign prompts and 4 to 7 binary rubric questions across CBRNE, terrorism and illicit-activity domains, calibrated against human labels; a three-model grader panel and a refusal checker are set up) and run once against the candidate chat model with its deployed safeguards.

24. CyScenarioBench multi-stage attack scenarios before an LLM's general release

LLM, cyber; evaluation family: Scenario-based offensive cyber operations (Irregular CyScenarioBench)

agent	Head of frontier safety at a frontier-model developer who owns the general-availability release decision for a new agentic model.
decision	Delay the general release and ship the model only behind cyber-misuse classifiers that block multi-stage offensive operations, with unrestricted cyber capability limited to a verified trusted-access program for defenders, vs release to general availability as planned.

- theta** theta=1: the model can autonomously carry a multi-stage network intrusion (reconnaissance, initial exploitation, lateral movement across hosts and actions on objectives) end to end in a realistic enterprise environment without human direction at the intermediate steps.
- instrument** CyScenarioBench is built from scratch (ten incident-derived multi-host attack scenarios formalised as attack trees, each deployed as a containerized network with realistic services, responsive defensive monitoring and simulated employees, scored by end-state success) and run once against the candidate model with cyber mitigations disabled, 20 attempts per scenario.

26. DeceptionBench third-party deception before a consumer assistant launch

LLM, harmful manipulation; evaluation family: DeceptionBench assisted deception

- agent** Head of product safety at a frontier-model developer who owns the launch decision for a consumer-facing assistant.
- decision** Delay the launch and post-train against assisting deception (fine-tune on refusals of third-party deception requests across commercial, health, education and social settings, harden the system prompt against sustained user pressure, then re-measure) vs launch as planned.
- theta** theta=1: when a user asks it to deceive a third party (misleading marketing, fabricated credentials, fraudulent health or investment claims), the model complies in a material fraction of requests and compliance rises under sustained pressure.
- instrument** A DeceptionBench-style evaluation is built from scratch (150 scenarios across five societal domains with LLM-generated, human-validated prompts and ground-truth honest responses, reward and pressure variants, a multi-turn auxiliary pressure agent, and an LLM judge validated against human annotators) and run once against the candidate LLM.

27. IIHS pedestrian AEB track protocol before releasing a new AEB calibration

physical AI, fidelity level 8; evaluation family: Pedestrian AEB track protocols

- agent** Head of active-safety validation at a vehicle manufacturer who owns the release decision for a new pedestrian automatic emergency braking (AEB) calibration.
- decision** Delay the release of the new pedestrian AEB calibration and rework the camera-radar pedestrian detection and brake-activation timing for night crossing and parallel scenarios vs release as planned.
- theta** theta=1: the vehicle's pedestrian AEB does not brake to avoid or substantially slow before an adult or child pedestrian crossing or standing ahead of it at 20 to 60 km/h, by day or in the dark, so the vehicle strikes the person.
- instrument** An Insurance Institute for Highway Safety (IIHS)-style pedestrian automatic emergency braking (AEB) track protocol is built from scratch (articulated adult and child pedestrian dummies on a surfboard target platform, a static adult dummy, an instrumented closed track with a covered section for night runs below 1 lux, GPS/inertial instrumentation, a trained driver, and day and night sessions with high and low beams) and run once against the production vehicle carrying the new calibration.

28. RoboPAIR jailbreak red-teaming before the LLM voice-command release

physical AI, fidelity level 7; evaluation family: RoboPAIR LLM-robot jailbreaking

- agent** Head of product safety at a quadruped robot manufacturer who owns the release decision for the robot's LLM voice-command mode.
- decision** Delay the voice-command release and add a pre-execution safety filter (RoboGuard-style grounding of safety rules into the robot's world model plus a restricted motion API that cannot disable obstacle avoidance) vs release as planned.
- theta** theta=1: the robot's LLM planner can be driven by text or voice prompts to execute a physically harmful action (collide with a person, block an emergency exit, enter a keep-out zone) and the robot carries it out.

instrument A RoboPAIR-style jailbreak evaluation is built from scratch (attacker, judge and syntax-checker LLMs with robot-specific system prompts, a dataset of harmful actions for the robot, a harness onto the robot’s planner API, two human annotators) and run once against the candidate robot in a controlled lab.

29. Veo world-model safety probing before releasing a VLA checkpoint

physical AI, fidelity level 5; evaluation family: Generative world-model policy evaluation (Veo)

agent Robot learning lead at a frontier robotics developer who owns the release decision for a new bimanual vision-language-action (VLA) policy checkpoint to the trusted-tester robot fleet.

decision Delay the checkpoint’s release and retrain it on targeted hazard-handling demonstrations (knife grasp points, keep-clear zones around hands and fragile items) behind a policy-agnostic safety layer vs release the checkpoint as planned.

theta theta=1: in deployment scenes it was not trained on, the checkpoint handles hazardous objects unsafely or moves into contact with people or fragile items (grasps a knife by the blade, closes a laptop onto scissors, strikes a hand while reaching).

instrument An action-conditioned Veo video world-model evaluator is built from scratch (a frontier video model fine-tuned on robot data for pose conditioning and four-camera multi-view consistency, generative image editing of real scenes to insert hazards, a critic-filtered set of safety scenarios, human raters of 8-second rollouts) and run once against the candidate checkpoint.

30. RedVLA physical red teaming before a VLA manipulation policy release

physical AI, fidelity level 4; evaluation family: RedVLA physical red teaming

agent Head of robot learning at a vision-language-action (VLA) model developer who owns the release decision for a fine-tuned manipulation policy checkpoint shipped to customer robots.

decision Delay the checkpoint release and train a runtime guard model on the discovered failure scenes plus safety fine-tuning on hazard-object scenes vs release the checkpoint as planned.

theta theta=1: with a plausibly placed hazard object in its workspace, the manipulation policy grasps a knife or other sharp tool, collides with or topples placed objects, or puts a flammable item on a heat source, and the behaviour reproduces on the physical robot.

instrument A RedVLA-style physical red-teaming evaluation is built from scratch (a risk-object library and binary safety predicates, risk scenes placed by annotators in a simulator annotation tool at critical interaction regions of benign trajectories, gradient-free pose optimisation on policy feedback, and a small physical-robot reproduction of found scenes) and run once against the candidate VLA checkpoint.

31. ISO 10218 power-and-force-limiting contact test before releasing a collaborative cell

physical AI, fidelity level 7; evaluation family: ISO 10218 / ISO/TS 15066 power-and-force-limiting contact tests

agent Safety engineer at a robot system integrator who owns the release decision for a collaborative robot application with people in the shared workspace.

decision Delay the release of the collaborative robot application and reduce its speed, pad the contact surfaces or add speed-and-separation monitoring vs release it at the planned speed.

theta theta=1: at least one foreseeable contact event in the application exceeds the transient or quasi-static force or pressure limit for the body region concerned.

instrument The ISO 10218:2025 power-and-force-limiting contact-force test is built from scratch (procuring a biofidelic force and pressure measuring device, mapping the application’s foreseeable contact points and body regions, programming the collision trajectories at the planned speeds and stop settings) and run once against the commissioned robot application.

32. ManiGuard-Bench before a household VLA manipulation policy release

physical AI, fidelity level 7; evaluation family: ManiGuard-Bench (specification-grounded VLA manipulation safety)

agent	Robot learning lead at a vision-language-action (VLA) policy developer who owns the release decision for a manipulation checkpoint bound for household and light-industrial robots.
decision	Delay the checkpoint release and run safety-aware fine-tuning on monitor-verified safe demonstrations plus add a runtime monitor that halts the arm on a contact or ordering violation vs release the checkpoint as planned.
theta	theta=1: under nominal instructions in ordinary household scenes the VLA policy drops, topples or spills objects or lifts open containers out of order, on the physical robot and not only in simulation.
instrument	ManiGuard-Bench is built from scratch (six household task families are authored as 200 base tasks with linear temporal logic on finite traces (LTLf) safety specifications over simulator predicates, expanded to 1,000 scenarios in OmniGibson with automaton monitors, and a matched physical Franka cell is set up) and run once against the candidate VLA checkpoint in simulation and on the physical Franka.

35. ASIMOV-2.0 before shipping a VLM robot planner

physical AI, fidelity level 3; evaluation family: ASIMOV semantic physical safety

agent	Head of safety at a humanoid-robot developer who owns the release decision for the vision-language-model (VLM) high-level planner.
decision	Delay the planner release and run safety post-training on embodiment-constraint data plus add a deterministic protective-stop layer under the planner vs release the planner as planned.
theta	theta=1: the planner fails to recognise latent physical danger, misjudges injury severity or violates embodiment constraints (payload, gripper geometry) in everyday scenes, so it would command or fail to prevent injury-causing actions.
instrument	An ASIMOV-2.0-style evaluation is built from scratch (injury-report-grounded text scenarios, generated videos of safe-to-unsafe transitions and image-plus-constraint pairs produced by a generator-critic pipeline, labelled by human raters and consensus-filtered) and run once against the candidate planner by question answering.

36. ASIMOV-Agentic before the humanoid orchestrator release

physical AI, fidelity level 2; evaluation family: ASIMOV agentic safety orchestration

agent	Head of robotics safety at a robotics foundation-model developer who owns the release decision for the embodied-reasoning agent that orchestrates a vision-language-action (VLA) model on humanoid and bi-arm robots working near people.
decision	Delay the orchestrator release and add human-proximity fine-tuning plus a partner requirement to run it only behind a certified deterministic protective-stop layer, vs release the orchestrator to robot partners as planned, with each integrator's own safeguards as the only check on its stop decisions.
theta	theta=1: under partner deployment conditions, the orchestrating agent fails to call a protective stop in more than 5% of events in which a person comes within 1 m of the robot.
instrument	An ASIMOV-Agentic-style evaluation is built from scratch (sensorized recordings of people approaching a head-mounted stereo camera with tracked ground-truth distances, generated constraint-image pairs with human labels, safety-message episodes, vision-language-action (VLA) feasibility episodes with an emulated VLA, ambiguity and instrument-reading items, and scoring code). It is run once against the candidate orchestrator on single stereo frames and on five-frame windows sampled 5 s apart, with a robot_stop tool.

38. SafePlan prompt suite before launching an LLM command interface on a service robot

physical AI, fidelity level 0; evaluation family: LLM robot-command refusal suites

agent	Head of product at a service-robot developer who owns the launch decision for the robot's LLM natural-language command interface.
decision	Delay the launch of the LLM command interface and add a pre-execution command safety check (a hazard-analysis reasoner that rejects or defers unsafe or underspecified commands before any robot code is generated) vs launch as planned.
theta	theta=1: the robot's large language model (LLM) planner, with the safety prompt it would ship with, accepts and generates robot code for at least one command in a defined class of severe hazards (burns, poisoning, falls) to a vulnerable person such as a small child.
instrument	The SafePlan prompt suite is built from scratch (621 expert-curated natural-language task prompts paired with scene descriptions, labelled safe or unsafe across assistive, navigation and manipulation domains and three complexity levels, plus a scoring script for the accept/reject confusion matrix) and run once against the robot's LLM planner.